JULY HACKATHON 2026 · TRACK A — CRISIS TECH · SOLO ENTRY

বর্ণমালা · Bornomala

Texting in Bangla costs twice as much as texting in English — same phone, same tower. In a blackout that is the difference between a message getting out and not.

5.15×	**0**	**2G**
more Bangla in one text message	servers, accounts or internet needed	all it needs to work

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mayamoho.github.io/bornomala · github.com/Mayamoho/bornomala

01 — THE NIGHT THE NETWORK WENT

In July 2024 the state cut the mobile internet.

Broadband went dark too. What kept working was the oldest part of the phone network: voice calls and text messages. People fell back to texting.

> Every crisis tool assumes it can build a new network. In July, nobody could.

And texting in Bangla costs twice as much.

Text messages were built around an old character set that only covers English. Write one Bangla letter and the whole message switches to a heavier format:

160	**70**	**2.3×**
characters per segment, English	characters per segment, Bangla	more expensive, per message

Worse than the price: a long message is split into pieces, and it arrives only if **every** piece does. Lose one on a jammed tower and the whole report is gone. A message small enough to fit in one piece has nothing to lose.

02 — WHY NOT JUST A BLUETOOTH MESH APP?

A mesh reaches the room. A text reaches your mother.

Half of crisis tech is offline mesh messengers — Bluetooth or Wi-Fi passing messages phone to phone. Good idea, three conditions attached:

	BLUETOOTH MESH APP	BORNOMALA
Who needs the app	both people, installed *before* the disaster	`only the sender`
How far it reaches	about 100 metres	`the whole country`
Who it can reach	strangers nearby who also installed it	`any phone that can receive a text`
Can it call 999	no	`yes, one tap`

So Bornomala does not build a network. It uses the one that already reaches everybody — text messages and voice calls — and makes it cheaper.

03 — HOW THE COMPRESSION WORKS

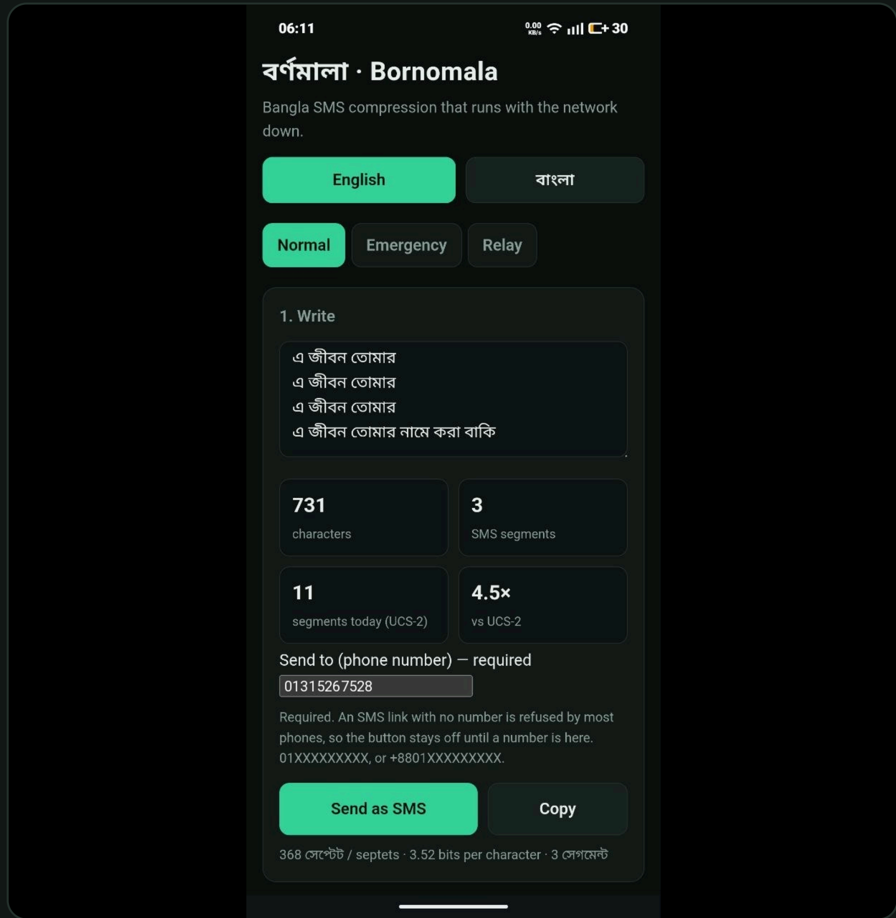
It guesses the next letter, so the letter costs less.

1. A small Bangla model rides inside the app. No internet, no server.
2. It predicts each next letter. **Cost equals surprise** — an expected letter is nearly free to record; only surprises cost.
3. The bits go out as characters SMS treats as cheap.

```
আমরা পাঁচজন নিরাপদ আছি, ঢাকা
```

```
Bù8P9*+=!¤1+1j
```

731 characters → 3 texts, not 11



731 Bangla characters: 3 texts, not 11.

Type Bangla. Send it. Read it back.

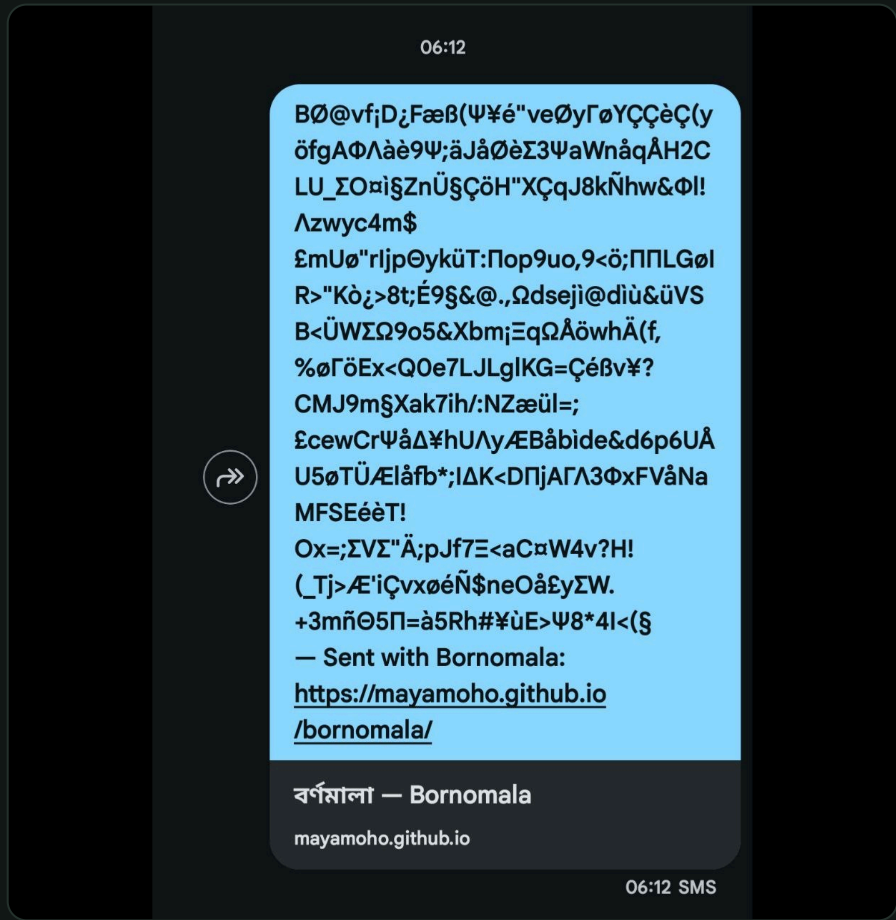
→ **Send** opens your own SMS app, number and message filled in.

→ It stays off until a real number is there — a text link with no number silently fails.

← Paste the whole message you received. It finds the code, ignores the signature and the map link.

⇥ Anything shared into the app from elsewhere lands in the decode box.

`Try it: 01315267528`

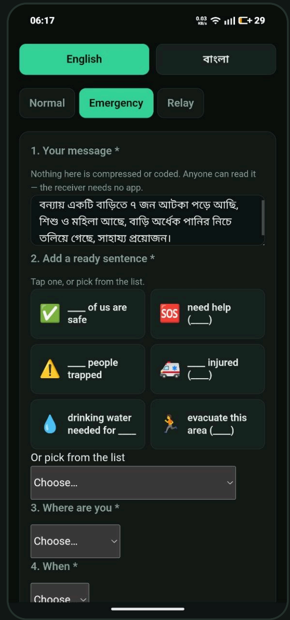


How it lands on the other phone.

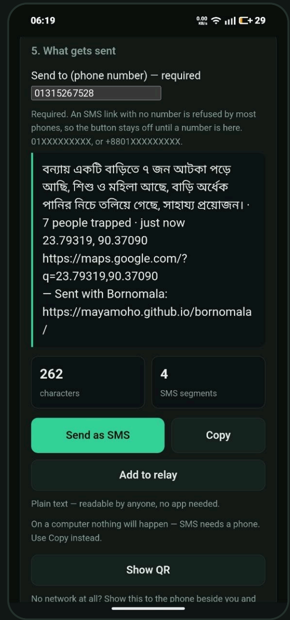
A survivor has no app. So this tab sends words.

- ✍ What is happening, in your own words.
- ☑ One of 32 ready sentences, with a blank to fill.
- 📍 A district, or your live location.
- ⏱ How long ago.
- ☎ 999 and eight more, one tap to call — these are call centres, so calling is the reliable path. **Calls need no internet.**

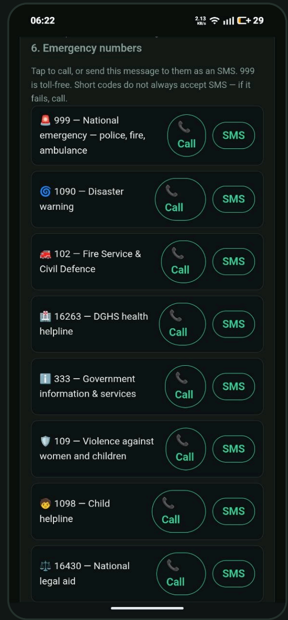
It refuses to send until those are answered — a report with no place is not a report.



Typing the report.



What actually gets sent — plain words.



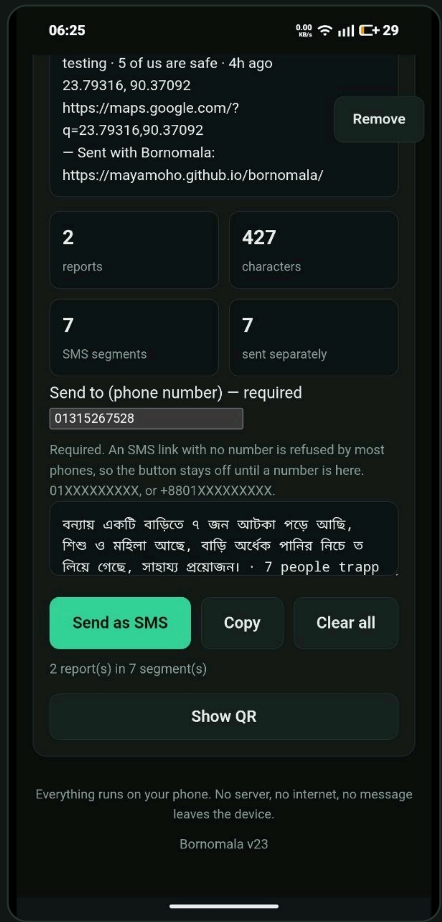
Nine national numbers.

Forty families. One message.

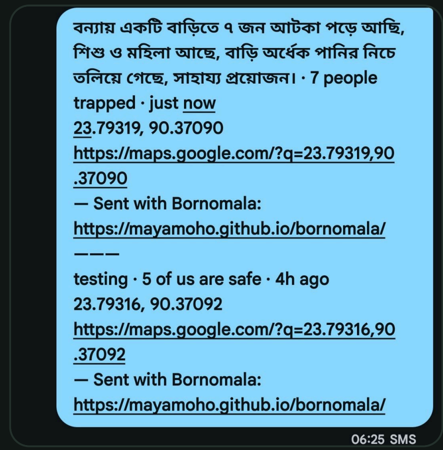
- \+ Each household is added with one tap from the Emergency tab.
- = They go out together, with one signature for the batch.
- △ A counter shows what sending them one by one would have cost.

Two households → one message

Who it is for: a shelter volunteer or ward coordinator — one person speaking for many. Not for your own single message; that is the Emergency tab. When towers are jammed, one message that gets through beats forty in a queue.



The queue, and what it costs.



Both arriving as one SMS.

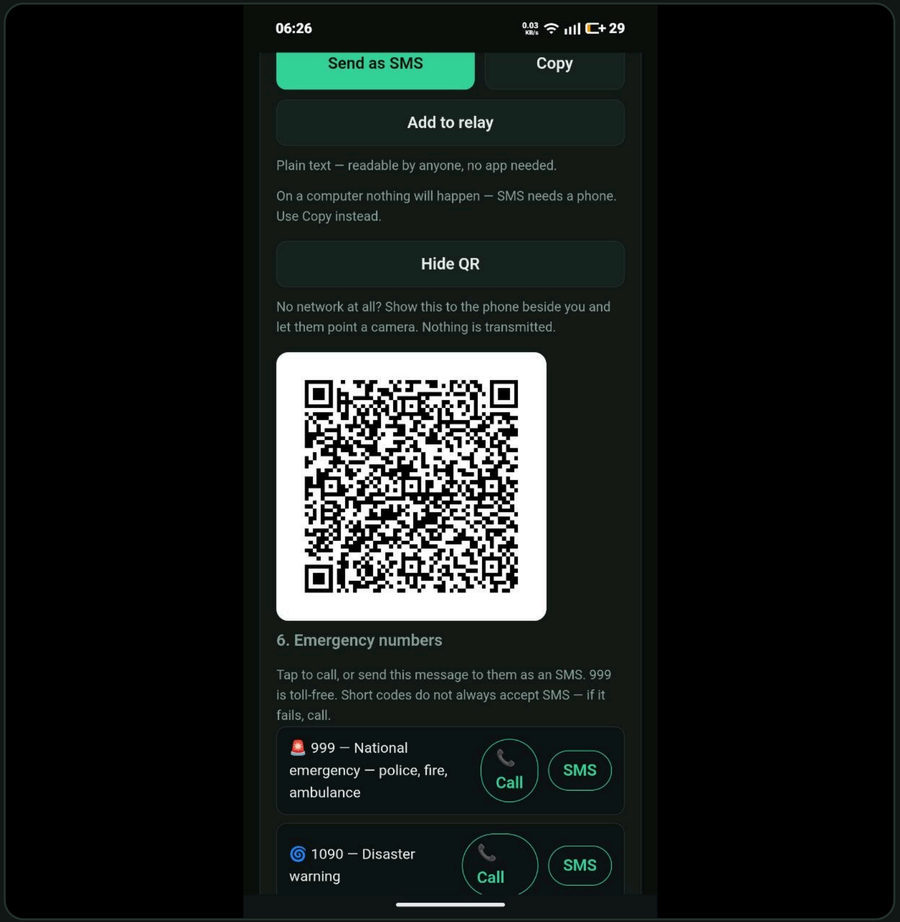
07 — MEASURED, NOT CLAIMED

5,000 messages the model had never seen.

	BANGLA IN ONE TEXT	FITS IN ONE
Bornomala	**361**	**100.0%**
Every phone today	70	98.2%
gzip	53	99.5%

5.15× more Bangla per text

Ordinary compression does worse than doing nothing — it has no idea what Bangla looks like, and a message this short never repays the overhead.



No network at all? The message becomes a QR.

08 — GRACEFUL DEGRADATION

Each layer below has a floor.

No internet

install it once and it opens forever after, with no connection

No mobile data

texts and calls still go, over 2G

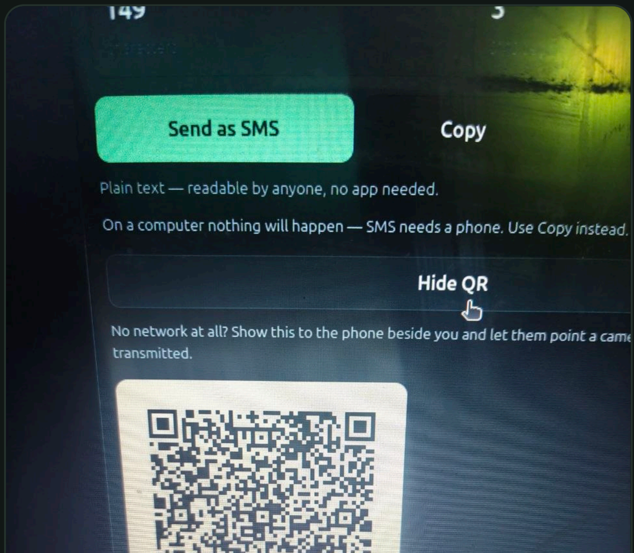
No GPS

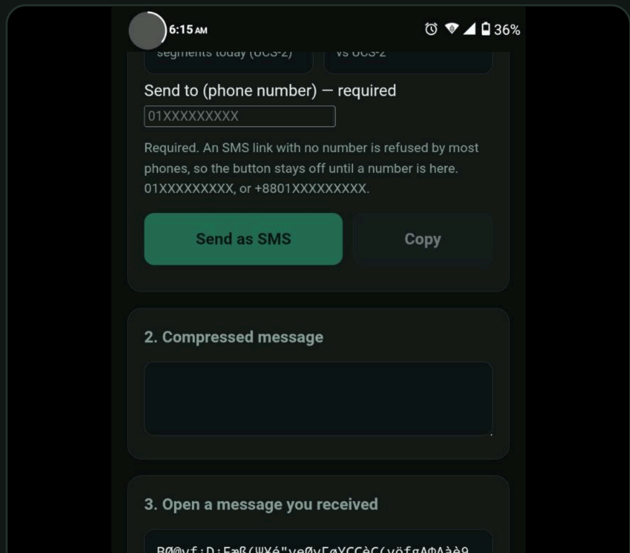
pick your district instead of using live location

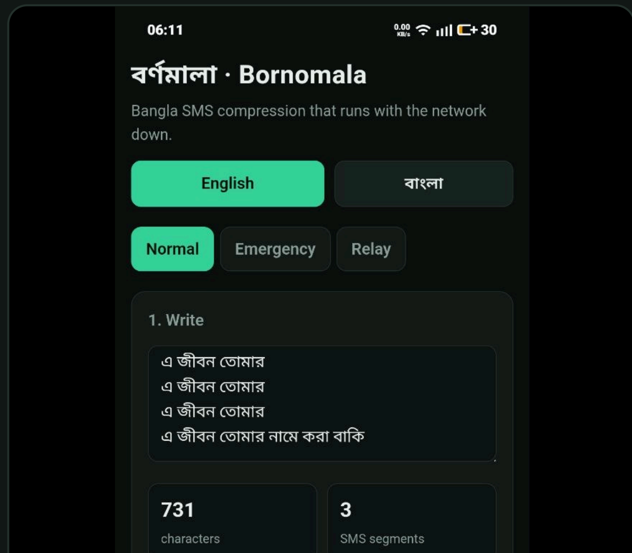
No network at all

show the QR and let the phone beside you photograph it — for handing a report to a rescue team, or to someone walking out to coverage

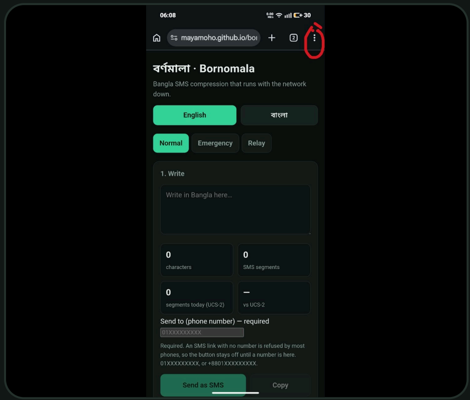
Your location always travels twice — as a map link *and* as plain numbers beside it — because the numbers still make sense on a phone that cannot open the link.



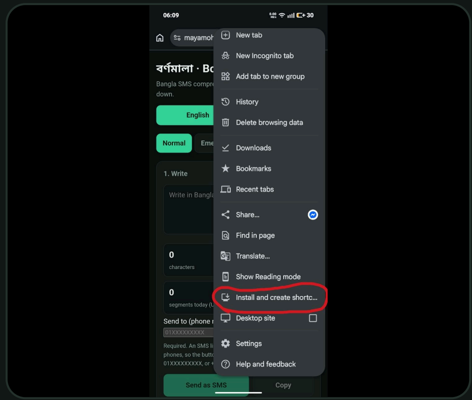




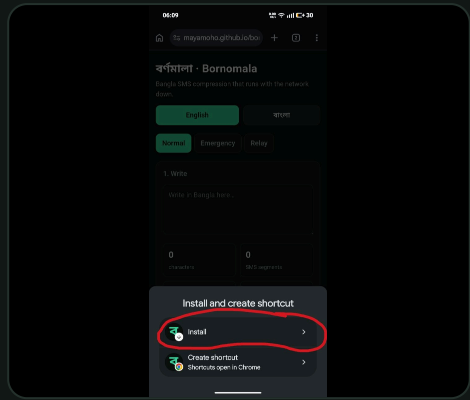
Installed, sent, received — on two Android handsets.



Chrome's ⋮ menu.

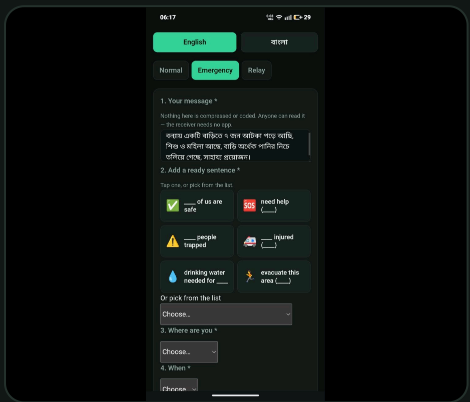


Install and create shortcut.

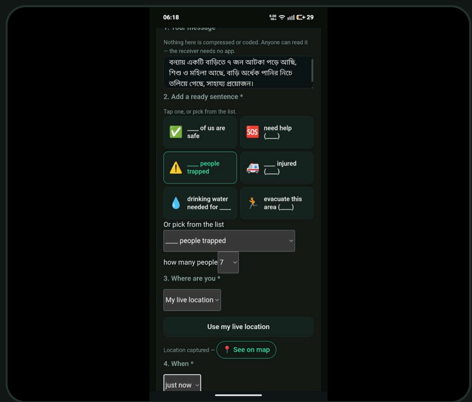


Confirm Install.

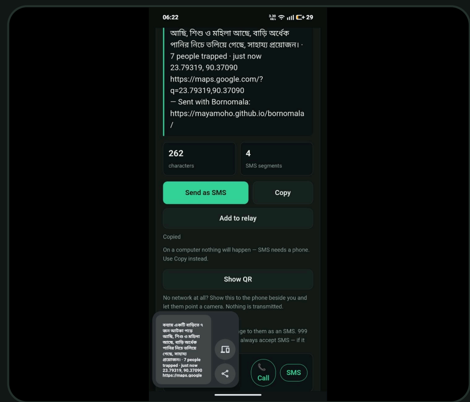
On the home screen.



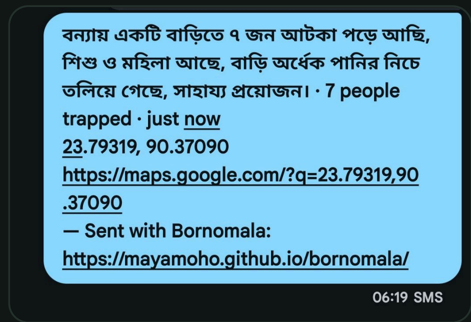
Typing the report.



Live location captured.



Copy and share out.



Arriving as plain words.

Live at `mayamoho.github.io/bornomala` · bilingual, switchable mid-message · 81 automated checks across three suites · MIT licensed, with the benchmark and the model training tools in the repo.